

XIN YANG, PH.D.

Boston, MA

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RESEARCH INTERESTS

Statistical machine learning for regulated, risk-bearing decisions; hidden Markov and mixture models; integrative (multi-unit) inference; calibrated and auditable predictive modeling for insurance, financial services, and pharmaceutical/health data.

EDUCATION

University of Connecticut, Storrs, CT, USA

Aug 2021 – May 2026

Ph.D. in Statistics Advisor: Dr. Kun Chen

Villanova University, Villanova, PA, USA

Aug 2019 – May 2021

M.S. in Applied Statistics

Nankai University, Tianjin, China

Aug 2015 – Jun 2019

B.S. in Statistics

PUBLICATIONS

Peer-reviewed journal articles

1. Nyiera, H. N., **Yang, X.**, Liu, Y., Cha, S., Chen, O., Hu, G., Chen, K., & Zhao, J. (2026). Correlating intensity fluctuations and lifetime distributions reveals surface chemistry effects on photoluminescence of single CsPbBr₃ quantum dots. *ACS Applied Materials & Interfaces*, 18(23), 32997–33006. doi:10.1021/acsami.6c03594
2. Nyiera, H. N., **Yang, X.**, Sun, Y., Wang, J., Liu, Y., Chen, O., Chen, K., & Zhao, J. (2025). An integrative statistical approach for analyzing photoluminescence fluctuations in single CsPbBr₃ quantum dots. *The Journal of Physical Chemistry C*, 129(38), 17150–17159. doi:10.1021/acs.jpcc.5c04795

Under revision

3. **Yang, X.**, Nyiera, H., Sun, Y., Zhao, J., & Chen, K. (2025). Integrative learning of quantum dot intensity fluctuations under excitation via tailored dynamic mixture modeling. arXiv:2501.01292. Under revision at *The Annals of Applied Statistics*.

Manuscripts in preparation

4. **Yang, X.**, Nyiera, H., Sun, Y., Zhao, J., & Chen, K. Beyond FLID: Joint modelling of fluorescence intensity and lifetime with mixture hidden Markov models.

RESEARCH EXPERIENCE

Joint Modeling of Fluorescence Intensity and Lifetime via Coupled Hidden Markov Models Sep 2024 – Present
University of Connecticut, with Dr. Kun Chen (Statistics) and Dr. Jing Zhao (Chemistry)

- Developed a coupled mixture HMM that jointly models intensity and lifetime trajectories of single quantum dots
- Integrative multi-dot inference that simultaneously estimates parameters and clusters particles
- Likelihood-ratio tests to formally assess dependence between the intensity and lifetime processes
- Research supported in part by NSF Grant CHE-2203854; collaboration continuing after graduation

Integrative Learning of Quantum Dot Intensity Fluctuations via Tailored Mixture HMMs Sep 2022 – Jan 2025
University of Connecticut, with Dr. Kun Chen

- Pooled information across 128 quantum dots to identify shared latent intensity states that single-particle analysis cannot recover
- Developed EM-based estimation algorithms and visualization of latent-state dynamics

Proportional Hazards Assumption in Cox Regression

Oct 2023 – Dec 2023

University of Connecticut, with Dr. Ming-Hui Chen

- Reviewed residual-based tests of the PH assumption; designed a simulation study comparing test performance

PROFESSIONAL EXPERIENCE

Associate Data Scientist

May 2026 – Present

Data Scientist Intern (Student)

Jan 2026 – Apr 2026

Plymouth Rock Assurance Corporation, Boston, MA

- Developing an AI-powered bodily injury claim model for the personal auto line to forecast ultimate loss, support accurate reserving, and improve claim operations
- Built an automated ETL pipeline using LLM APIs to parse unstructured claim photos, adjuster notes, and questionnaires into structured features, combined with ten years of historical claim data
- Built staged submodels that flag high-exposure claims early in the claim lifecycle
- Deploy version-controlled, production-grade Python on AWS SageMaker for repeatable, monitored model runs
- Work with claim adjusters, the claims director, and actuarial to align model outputs with reserving practice

Data Scientist Intern

May 2024 – Aug 2024

Plymouth Rock Assurance Corporation, Boston, MA

- Evaluated external data for catastrophe claim modeling with XGBoost; assessed calibration and proposed a refined flood-risk feature (Distance to Flood Zone)

CMC Statistician (Co-op)

Jan 2021 – May 2021

GlaxoSmithKline (GSK), Collegeville, PA

- Provided CMC statistical support for bioassay transfer and method validation in new product development
- Estimated HCP and DNA residual thresholds with bootstrap and nonparametric methods
- Assessed precision and accuracy with TOST equivalence testing per ICH Q2(B); reviewed protocols for regulatory compliance

HONORS & AWARDS

- NESS Student Paper Award, New England Statistics Symposium (2025)
- ICSA Travel Award, International Chinese Statistical Association (2025)

PRESENTATIONS

- New England Statistics Symposium (NESS), 2025
- ICSA Applied Statistics Symposium, 2025

PROFESSIONAL SERVICE & MEMBERSHIPS

- Member, New England Statistics Society (NESS)
- Member, American Statistical Association (ASA)

TEACHING

Teaching Assistant, Introduction to Statistics (STAT 1000Q)

Fall 2023

University of Connecticut – weekly sessions for ~80 students; office hours and exam reviews

SKILLS

Statistical Modeling:	Hidden Markov models, mixture models, EM algorithm, survival analysis, bootstrap & nonparametrics, equivalence testing
Machine Learning:	XGBoost, LightGBM, random forest, LLM-based information extraction
Programming & Cloud:	R, Python, SAS, SQL; AWS SageMaker, Snowflake; Git, $\LaTeX$